6T SRAM Bitcell Characterization

Transistor-Level Memory Design and Stability Analysis

Project Overview

This project details the transistor-level design and characterization of a standard 6-Transistor (6T) SRAM bitcell using LTspice. The core objective was to balance cell area with operational stability. The circuit features cross-coupled CMOS inverters for data storage and two NMOS access transistors controlled by a Wordline (WL) to interface with the Bitlines (BL, BLB).

> **Design Methodology:** Proper transistor sizing (Cell Ratio and Pull-Up Ratio) is critical to prevent destructive reads and ensure reliable writes. The PMOS pull-up network, NMOS pull-down network, and access transistors were iteratively sized to optimize the Static Noise Margin (SNM).

Transistor Sizing (W/L)

Transistor Type	Role	Width (W)	Length (L)
M1, M3 (NMOS)	Driver / Pull-Down	120 nm	45 nm
M2, M4 (PMOS)	Load / Pull-Up	60 nm	45 nm
M5, M6 (NMOS)	Access Transistors	90 nm	45 nm
Cell Ratio (CR)	W_driver / W_access	1.33	

Performance Analysis

Transient analysis was performed to measure the Read and Write access delays under loaded bitline conditions. Stability was quantified by generating a Butterfly Curve using a DC sweep, allowing for the graphical measurement of the Static Noise Margin (SNM) during standby and read operations.

READ SNM	WRITE DELAY	TECH NODE
145 mV	**28 ps**	**45 nm**